8. Diagram (equipment to be used is shown in the diagram)

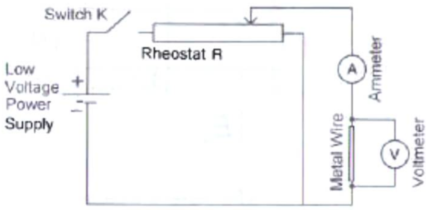

Fig. 8.1

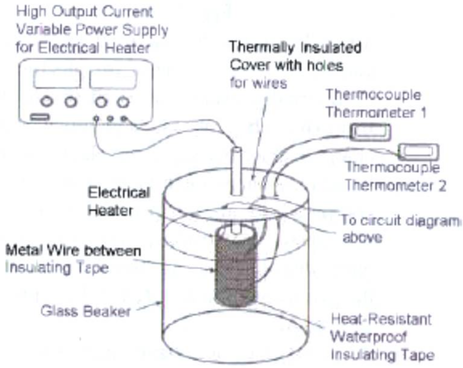

Fig. 8.2

Aim: To determine the value of α for different metals in the temperature range 0 °C to 200 °C.

Independent variable: type of metal wire, temperature θ

Dependent variable: Potential difference V and current I.

Procedure:

(a) Refer to Fig. 8.2. Wrap an electrical heater with heat-resistant waterproof insulating tape. Wind the test wire around the electrical heater with the probes of the two thermocouples at the two ends of the test wire. Wrap another layer of the same tape around the test wire.
(b) Suspend the heater, test wire and thermocouple thermometer probes in the middle of the glass beaker, from a thermally insulated cover.
(c) Initially, with the heater turned off, fill the beaker with crushed ice and water. Let the system of the heater in the ice-water bath stabilise.
(d) Record the average temperature θ of the test wire from the readings of the two thermocouple thermometers.
(e) With switch K closed (Fig. 8.1), the rheostat can be adjusted to record corresponding readings of V and I. Obtain at least 6 sets of V and I readings. The resistance R of the wire at this temperature can be determined from the ratio of V to I. As such the average resistance of the metal wire at this particular temperature can be calculated.
(f) Turn on the heater and repeat steps (d) and (e) for temperatures θ up to 200 °C, without the ice water. There should be readings of V and I taken for at least 6 different temperatures. As such, obtain the resistance R of the wire at various temperatures θ. Tabulate the respective data collected.
(g) Plot the graph of R against θ. R_o is the resistance of the wire at 0 °C and can be obtained from the y-intercept. The gradient of the graph is αR_o. Hence the constant α for this particular metal wire can be calculated.
(h) Change a test wire of another metal type. Repeat steps (a) – (g) to obtain the constant α for different metals.

Precautions:

(a) Another set of readings can be obtained while the wire is cooled from 200 °C to 0 °C so that the average can be obtained to minimize the effect of random errors.
(b) Ensure that the wire is properly separated from the heater surface by the insulating tape and water does not get in between the tape so as to prevent the test wire from being short-circuited.

Safety:

As the heater operates at high temperatures, it can cause severe burns. Hence, it is suggested that the experimenter wears thermally insulated gloves to prevent burn injuries.